

Ujjwal Agrawal

Software Engineer 2

ujjwalagrawal3012@gmail.com | +91-8763343874

[Portfolio](#) | [LinkedIn](#)

EDUCATION

- Ramaiah Institute of Technology | Aug '18 – Jul '22 | Bachelor of Engineering | 8.6/10

EXPERIENCE

Software Engineer 2 - Vyapar TaxOne (Formerly Suvit) | April '26 – Currently

- **Built a Tally connection framework that auto detects free ports and rebinds per-user Tally instances**, fixing a contention bug where only the first Tally bound and listened while every other instance's ODBC connection silently failed, unblocking concurrent multi-user access.
- **Led the migration of the primary data store from MongoDB to a sharded PostgreSQL + Citus architecture** designed for horizontal scale, with shard-key strategy, distribution tuning, indexing, and failover.

Software Engineer 2 - Simply Vyapar Apps | July '24 – Currently

- **Reduced database queries by 70% and improved page load performance by 3x (2.1s → 0.7s)** by architecting a sync-aware, DB transaction aligned hybrid cache invalidation system that replaced inefficient full-reload patterns with selective refresh logic.
- **Eliminated 100% of partial commits and reduced transaction-related defects by 85%** by designing and implementing a distributed transaction framework enforcing atomic commits between local and server databases (commit-on-remote-success model).
- **Reduced PDF-related support tickets by 75% and eliminated non-recoverable production failures** by designing a resilient PDF generation pipeline with multi-engine fallback (native print API → headless browser).
- **Reduced HTML payload size by 66% and improved preview stability/performance by 50%** by redesigning the invoice/PDF generation framework in an Electron desktop app, replacing DOM-bound rendering with IPC-based file-backed previews.
- **Standardized UI architecture across 10+ business modules** by architecting a reusable composite page-layout framework (React + TypeScript) with modular, typed sections (titular, stats, filters, table, detail, common states), cutting per-page development effort by 25% and eliminating duplicate frontend code via plug-and-play integration.
- **Eliminated full table scans on large datasets** by building a two-phase Master Search, a pre-built FTS3 index resolves matches across 9+ fields in $O(\log n)$ time, and only matched IDs are passed to a targeted paginated query with date, type, and role filters.
- **Prevented UI crashes on large datasets** by engineering a bidirectional infinite scroll model that keeps only 70 rows in memory at any time, evicting old blocks on forward scroll and re-fetching on backward scroll, with reactive datasource resets ensuring data stays fresh on every filter, sort, or search change, and full parity maintained across Excel and print exports.
- **Improved dashboard responsiveness by 40% and reduced unnecessary re-renders by 30%** by architecting a real-time dashboard framework with centralized state management, enabling dynamic widget selection and multi-range sales analytics.
- **Improved user activation by 30% and reduced onboarding drop-off by 15%** by building a Firebase Remote Config-driven A/B experimentation system that split first-time users between a demo-scheduling flow and a guided first-sale walkthrough, with phone/OTP authentication and a skippable gate controlling entry.
- **Built role-based item visibility controls for salesman management**: admins can now restrict salesmen to specific item categories, with smart defaulting logic ensuring newly added categories automatically inherit the correct access without any manual reassignment, backed by DB schema changes and a redesigned admin UI.

Frontend Developer - BizAcuity Solutions | July '22 – July '24

- **Improved development velocity by 20%** by leading the frontend team in migrating the codebase to React.js and TypeScript, standardizing component architecture across the application.
- **Drove 10% business growth** by building a bulk user onboarding feature enabling CSV-based imports, significantly reducing manual setup time and improving customer adoption.
- **Increased user conversion by 30%** by redesigning the website signup and login flow, implementing secure authentication, optimized validation, and friction-reducing UX improvements.
- **Reduced incident response time by 30%** by building an event-driven alerting system with automated email notifications and centralized alert tracking.
- **Reduced page load time by 30%** by translating UI/UX designs into optimized frontend implementations and applying performance optimization strategies to improve user experience.

ACHIEVEMENT

- **Won Simply Vyapar Tech Hackathon 2025 (Generative AI track)**, competing against internal engineering teams to build an AI-powered solution judged on technical depth and business impact.

SKILLS

- **Frontend**: React.js, TypeScript, JavaScript ES6+, HTML5, CSS3, Redux/Redux Toolkit, Formik, AG Grid, Firebase, Material UI, Bootstrap, SASS, Radix, Electron
- **Backend**: Java, Spring Boot, REST APIs, Node, Express, WebSockets
- **Databases**: SQL, MySQL, Vertica, Postgres, Mongo
- **Architecture**: Distributed Systems, Transaction Management, Caching, State Management, ACID
- **Analytics**: Mixpanel, Clevertap, Sentry
- **Tools**: Git, Figma, Jira